

Minimum System Requirements

The following components are the minimum configuration needed to build the **Vertical Form Fill Seal Machine V1.0**.

Hardware: Components such as drives, motors, S7-1200 G2 model or HMI model **can be changed** according to requirements.

Software: License types **can be tailored** to your specific needs, **ensuring good operation** of the machine.

BOM - Vertical Form Fill Seal Machine V1.0

Hardware requirements

Units	Component	Short designation	Article Number	Firmware version
1	PLC SIMATIC S7-1200 G2	CPU 1214C DC/DC/Rly	6ES7 214-1HH50-0XB0	V1.0
1	Analog Input Module	SM 1231 AI8	6ES7 231-4HF50-0XB0	V1.0
1	Unified Basic Panel	MTP1200 Unified Basic	6AV2 123-3MB32-0AW0	Not Applicable
1	Servo Cross Sealer Lift Axis	SINAMICS S200 PN	6SL5510-1BB10-1AF0	V6.3
1	Servo Filling Axis	SINAMICS S200 PN	6SL5510-1BB10-1AF0	V6.3
1	Servo Foil Feed Axis	SINAMICS S200 PN	6SL5510-1BB10-1AF0	V6.3
1	Synchronous motor Cross Sealer Lift Axis	1FL2 Synchronous motor	1FL2102-2AG01-1MC0	Not Applicable
1	Synchronous motor Filling Axis	1FL2 Synchronous motor	1FL2102-2AG01-1SC0	Not Applicable
1	Synchronous motor Foil Feed Axis	1FL2 synchronous motor	1FL2102-2AG01-1MC0	Not Applicable
3	Motion connection cables	MOTION-CONNECT 350 Power cable	6FX3502-7CD01-1AF0	Not Applicable

BOM - Vertical Form Fill Seal Machine V1.0

Software requirements

SIMATIC STEP 7 V20 Engineering Software

Component	License	Article Number Download	Article Number Package
TIA Portal V20 + SIMATIC STEP 7 V20 Engineering Software	SIMATIC STEP 7 Basic V20 Floating License	6ES7822-0AE24-0YA5	6ES7822-0AA24-0YA5

SIMATIC WinCC Unified Engineering V20

Component	License	Article Number Download	Article Number Package
SIMATIC WinCC Unified Engineering System Base Packages	WinCC Unified Basic ES V20	6AV2151-0XA02-4LA5	6AV2151-0XA02-4AA5

SINAMICS Startdrive V20 Engineering Software

Component	License	Article Number Download	Article Number Package
SINAMICS Startdrive Basic V20	Startdrive Basic V20	6SL3072-4LA02-0XG0	Not Applicable

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Libraries types

The libraries listed below are already contained in the Global Library **Automation Framework V2.2**

<https://support.industry.siemens.com/cs/document/109817223>

Component	Version	SIOS Link
Axis Control (LAxisCtrl)	V1.4.0	https://support.industry.siemens.com/cs/document/109925893
Library of Basic Controls (LBC)	V 3.1.0	https://support.industry.siemens.com/cs/document/109792175
Libraries of PLC data types (LPD)	V 1.0.0	https://support.industry.siemens.com/cs/document/109482396
Library of general functions (LGF)	V 5.1.1	https://support.industry.siemens.com/cs/document/109479728
SIMATIC OMAC PackML V2022 Mode&State Management and Machine Data Interface (LPMLV2022)	V 1.0.0	https://support.industry.siemens.com/cs/document/109821198
Lsicar	V 5.1	https://support.industry.siemens.com/cs/document/109804254
Library for SNTP Server Functionality (LSNTP)	V 4.0.0	https://support.industry.siemens.com/cs/document/82203451
Library of Unit Control - LUC	V 2.2.0	https://support.industry.siemens.com/cs/document/109974940

The following libraries are included by default. You can customize the selection, adding or removing libraries as needed.

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